

# Lance Da Costa ~ Active Inference Insights 010 ~ Mathematics, Time, Selfhood

*Insights | Active Inference Insights 010 ~ Mathematics, Time, Selfhood | 1:30:15*

hello everyone and welcome back to  
active inference insights as always I'm  
your host darus parvey Wayne and today I  
have the great pleasure of speaking to  
Mr although very soon it will be Dr  
Lance D Costa Lance is a PhD student  
with Greg Pavus and KL Friston jointly  
at Imperial College London in UCL and  
his work lies at the intersection of  
mathematics cognitive science and  
machine learning indeed he is at the  
very Cutting Edge of the technical  
formulations underlying AC of inference  
and so I'm personally looking forward to  
learning from him today and I hope you  
are too ladies and gentlemen I bring you  
Lance the Costa Lance thank you so much  
I I genuinely meant what I don't really  
like doing the introduction because it  
feels so you know sort of a bit stagnant  
and rigid but I genuinely meant what I  
wrote there which is that I'm really  
looking forward to learning than for  
having me and appreciate the  
introduction yeah um and you will when  
when is when is one when do you expect  
to become a doctor when's that total  
happening so my defense is at the end of  
February so oh my gosh it's inth yeah  
wow fingers crossed I'm sure you'll  
smash it thanks so before we start  
getting into the maths I this is very  
exciting for me because you're the first  
I would say pretty much pure  
mathematician we've had on we've had

computer scientists we've had cognitive scientists but no one really uh delving into the the nether regions of the maths um you know when I was researching what you're interested in I came across um the fact that you're interested in World models and I'm also very interested in World models so I wanted to start there as's like a teaser before we get into the maths and I wanted to ask whether you could explain what you mean by a world model and how you distinguish that from a self model and the reason why I ask that question is kind of twofold one is that I think the word world models might be slightly misleading because it kind of implies that we're modeling the world and in some sense we are but we're definitely not modeling the entire world so it might be called something like an econ Niche

model yeah although that's less catchy and then the other reason is I think there's still an open question at least to my eyes in active inference about whether we are modeling the world and ourselves or whether we're just modeling the world and we fit into that in some way right so it would be great to unpick that if possible yeah that that's actually a Cutting Edge question but I really like it great so let me start by world

what I mean what I personally mean by World model is that you as an individual you have always some sensory data coming in off active Visual auditive and so forth and you're going to build a model

for what's going on outside that causes your data um now the model that you build is a generative model which means that you can generate kind of factual data about what might happen in the future what might happen if I do this and that uh it's a probabilistic model because you're always uncertain about what's going on around in the world and it's also a causal model um this means that you're not um you're not exploiting the well you're not exploiting just the correlations in the data that come in but you're actually explaining the causes of the data something like more powerful than than a statistical model uh now I think you're right in calling it an eon Niche model because I mean the data that we have coming in at any point in time is just from around us and so that's ultimately what we want to model uh when we talk about the free energy principle we say that the brain optimizes its model to maximize accuracy and minimize complexity um in that and that entails uh having a model that's as simple as possible and therefore you don't want a model things are like too far away from you important from you and so this is why having an eon model as a terminology makes more sense sure so when you talked about world model and self model and this is where it really gets Cutting Edge uh so in the like simple applications of active inference and AI you just care about modeling the world um but now as humans we don't only

do that so for instance if I take a particular if I make a particular decision or take a particular course of action and you ask me why did you do that I'm not going to be able to uh answer in saying oh this neuron did this and this neuron did that and this happened but I'm going to give you a high level overview using natural language of why I did this now this uh is the idea of metacognition which means of having a model of the self in addition of having a model of the world so you have a model of yourself and you have a model of your own cognition and so this is kind of like a higher form of cognition and active inference uh which has not so far been really implemented out there I believe at least in machine learning it's kind of come out in papers on um modeling meditation with active inference L's work yeah I like yeah yeah yeah very nice cool that's really so so you so this is yeah this is a question I've been entertaining when I wrote this first paper on Flow that I've now mentioned far too many times so I shall mention it again um I was thinking exactly what are we talking about when we're talking about something like an epistemic agent model and uh I remember one comment commentator said oh is it just metacognition so that's an interesting alignment there um because in a sense yes you're right we have to reflect on our own capacities and model that as an

inference my only question there is you know in L's paper you have these really beautiful hierarchical Bayes graphs do you think there's a there has to be an upper bound on the computational complexity of that metacognitive and what and is is that just a limitation of the computation in and of itself or does it mean that there might be something ontologically sort of actually there right what what do you mean by ontological exactly uh let's well let's let's use a word well I'm not going to invoke something like a Cartesian soul but is there any I guess is there anything unique beyond the algorithm beyond the probabilistic model right uh so what I would say to that I mean what we're personally investigating with Lars is what does metacognition actually allow us to do allow us to do that's uh you know important for survival that an agent that does not have metacognition is not able to do so the idea is if metacognition arose at some point is because it increases our chances of survival our fitness it minimizes our free energy um how exactly I mean how is metacognition useful beyond what you know what we just talked about uh is currently unknown at least for me sure yeah I think it's I think it's a complex issue because it it's entangled with common folk psychology Notions of what it is to be a self um so most people will think of themselves uh probably if you ask them they would say yes I am this substantial thing somewhat

behind my eyes uh that persist through time so somewhat of some kind of cartesianified ego and I actually found I find that active influence is a lot easier to pass once we take that out of the equation when you just have the physics or when you just have the statistical mechanics it becomes a lot easier to say okay maybe this is maybe the self is just a further inference that's kind of where I was pushing you in terms of the ontological thing because I think people listening if they haven't really you know thought about the the Buddhism and the metacognition and the Derek Parfit related philosophy might think well there's just something there that's doing it right there's something there that's a homunculus behind the eyes that's actually doing the calculations that's doing the variational basing inference but I assume you can't you know there's no notion of that in your models um so you mean like kind of like an agent inside the agent indeed and obviously this leads to a homuncular infinite regress because who's watching the screen and so you know and so on but I think people have an issue with just a raw notion that there's just algorithmic computation going on because they're going to ask well who wait what who is doing it and that leads us back to some kind of uh active agent that's actually there not just an emergent phenomenon right uh so so I have a few things to say about that I think uh so

when we when we engage in meditation we are computationally looks like we're building a higher level in our giant model which is kind of like another age inside that makes inferences about our cognition yes and so when you're a trained meditator you not only think about the world or experience the world but you experience your own cognitive processes and you train yourself in doing that now how that looks like computationally is that you have a hierarchical gen model with many levels of of abstraction but when you translate that into math and physics it's kind of this idea of the Russian do yes like an outside agent and some more like more agents inside hierarch stacked that just infer each other

yes Mark of blanket in some sense yeah nested Mark of blankets and so but that but you know that analogy implies that there is some outer

shell that might be right only that might be you know yeah this makes me think of Maxwell Ramsey the M's paper on Consciousness that there's this right only layer that is contextualizing but not

contextualized um but that's kind of the other way around right that's like that's the that's kind of like the prime mover unmoved whereas your I feel like your argument is saying well everything is contextualizing everything else and actually what we might call the AL shell the autobiographical self is just the aggregation of all of these other selves

does that sound about right yeah it I  
feel it's uh kind of hard for me to to  
describe it um and I'm also thinking  
along these lines but I would say I mean  
the freeer principle everything is  
dependent on the blanket that you select  
right and is the idea that we have  
multiple nested blankets now the story  
that you tell depend on which blanket  
you decide to talk  
about do you I think people have an  
intuitive folk psychology notion that of  
course there's a point at which the  
modeling of me as a discrete identity  
needs to stop St so we could keep Mo so  
so most people probably think about the  
surface of the skin the epithelium and  
say well that that that's me right and  
then we can get into conversations about  
extended mind hypothesis but I actually  
don't think it's that relevant here I  
guess the point is  
um it seems like that the point at which  
we stop the mark of blanket is actually  
it's not arbitrary but it's an open  
question it's philosophically vague so  
to speak um which which which negates  
the kind of folk psychology notion that  
there is something like me which has  
these strict  
bounds yeah so so I I completely agree  
with kind of like the dissolution of  
those straight bounds once you start  
looking at it very carefully intuitively  
I think that my you know my hands are  
mine because I can control them in a  
relatively precise way and I think this  
is where that comes from okay it is the

preciseness at which you can act on something or inert that's nice

yeah when you when you do a lot of meditation or when you when you look at it like very carefully you start realizing that there's so many things in your body that you cannot control it's kind of obvious when when you think about it and you start realizing that you have very little agency over many things that are happening in your body so I think everyone of us has things that they're satisfied and they satisfied about themselves when you think about it where maybe I'm dissatisfied with something about myself uh but then this is not really my fault of me being that way so here's an example of um of something that I cannot control once you reflect that on that more and more you start realizing that you know uh we are self-organizing systems just like a tornado is a self-organizing system and we have a lot of things that are going on inside our bodies inside our minds that you cannot just cannot control H so yeah that's interesting I

guess I guess my rebuttal to that would be arguably the literature says that or suggests there are processes that underly something like a uh minimal self like inter receptive inference so that can be done reflexively or through kind of alistic control so let's say my glucose level drops Beyond a certain level I either you know secrete the

necessary hormones or go out there and  
eat but let's just say I'm secreting  
those hormones I'm not act you know my  
autobiographical self doesn't have the  
doesn't have the sense it's doing  
anything yeah th that mechanism of  
prediction Eros suppression is  
supposedly at the heart of the sense of  
mindness and  
presence so how do we resolve that if  
we're casting selfhood as that which is  
within my  
control um well I I don't know to be  
honest  
but it's a it's a a difficult question  
but  
uh yeah I mean the the more the more I  
think about all of this and the more I  
read the more I realize that we have  
control over very little yeah  
and so in in the meditation books so I  
keep coming to that because they're a  
source of inspiration for me for  
describing high level cognition there is  
this um like fundamental truth qu quote  
unquote which that there is no self the  
more you start like investigating the  
Mind through self inquiry the more you  
realize that there is no self yeah yeah  
um so maybe it speaks kind of to the  
things that we're just discussing sure  
yes this is yeah this is an interesting  
point um I guess you one would have to  
ask very dedicated meditators uh whether  
when they're doing let's say Vasa and  
they break that kind of Duality um  
because you know the subject OB object  
duality yielding to Oneness is kind of a

seminal phenomenon that occurs whether that sense of mindness and presence is also lost so there's an you know uh John vvi who's one of my favorite cognitive scientists and who was on the podcast speaks about adverbial qualia and what he means by that is you there are these so-called pure Consciousness events and even in those the sense of presence and perhaps mindness although we kind of weren't sure um might still be present which I which to my ears sounds like some computational mechanism like interceptive inference still being at play so I I guess this is calling for calling for a denial of the strict distinction between the body and the mind that the interceptive inference can still underly or bodily inference can still underly very important phenomenology yeah well uh active inference is definitely taking this uh philosophical position that cognition is embodied so so yeah it speaks exactly to what you said now I I just want to add something else so we talked about the the model of the self and I would say there's actually two models of the self so there's the model of the body and and this has a like clear um evolutionary like reason for being there so for example if something in my body hurts I need to figure out what to do about it for it not to hurt so it's about preserving your homeostasis and fulfilling preferences now there's this other level that we've also been discussing which is like the model of

our own cognition these are two  
different things yes that's that's  
interesting yeah cool there's lots  
there's lots to chew on there I'm glad  
that I'm glad that this is Cutting Edge  
questions that need Cutting Edge these  
are really things uh that I know a lot  
of people are thinking about spend  
yeah I think I think I'm gonna I'm gonna  
have lws on at some point  
yeah he owes me that to be honest uh he  
keeps promising okay um let's to the bet  
you know this is this self-modeling  
philosophical conversation is more my  
bag but let's take that deep PL uh  
plunge into the mathematics so as a as a  
kind of a forewarning or a precursory  
warning to the audience I'm not a  
mathematician um so everything I'm doing  
when I'm reading the math I'm trying to  
ground in intuitive philosophically  
robust Concepts so just just for the  
audience and yourself please do excuse  
me if I [ ] up um you won't I mean this  
is a I think it's very important to be  
able to talk about these things without  
the  
math it's literally impossible to talk  
in math so yes yeah yeah yeah there's  
this about this in a in a natural  
language way that's hopefully as simple  
as possible please well yeah I guess  
that's how it has to be taught is  
through metaphor um sort of yeah life  
through metaphor for George Lov kind of  
stuff so um start starting right at the  
basis in Act of inference the world is  
described as a random dynamical

system defined or described by a stochastic differential equation like a Langevin equation yeah why why is it worthwhile starting there why is that the starting point right um should I explain briefly what a Langevin equation is perhaps sure it'll be better you doing it than me okay uh sure well um so here's the thing we want a fundamental theory of the brain and cognition and we want it to be grounded in math and physics um now if you look at the fundamental theories that exist in physics these would be uh general relativity classical mechanics statistical mechanics quantum physics they are all based on stochastic differential equations or Langevin equations at some level now um a stochastic differential equation is a very fundamental and also basic object and I'll explain why it's basic it's basic in the sense that when you're modeling a system there are the known knowns and they're the known unknowns yeah stochastic differential equation uh brings the two together so there may be the known knowns about the system which could be forces that act on different components of the system this is going to drive the system deterministically along a predetermined path and then there are known unknowns these could be random fluctuations for example if you have a drone this could be random fluctuations due to wind if you have a boat random fluctuations due to waves

and these are things that you possibly cannot accurately capture so you're going to summarize them in a random term and then together so you have this deterministic uh term that gives you a predetermined traj and you have this random term that sort of perturbs the trajectory in a random way and when you put the two together you get kind of like a random evolution of your system with some structure and this is basically what the Lan equation or stochastic differential equation um encodes so this is to say uh it's a very general starting point and it's also the starting point that's consistent with the rest of physics and that's why we start there when we uh when we think about the free principle

excellent yeah so there's a couple things I wanted to pick up um am I right to say that it's about the rate of change of so of some  $X$  let's say of some states is that is that so that's that's kind of what we're looking at here is the okay and then the second thing is um this notion of random noise if you speak to certain philosophers for example they might say well it depends on who you ask I guess um because for a universal you know for um for some Demon that knows the entire unfoldings of the world in a single glance there's no Randomness true and yet to us the modelers well it's just computationally impossible to know every single element of the wind let's say or

the waves so what what are we talking about are we talking about Randomness in in and of itself in the actual system that's being modeled or for the

modular this is a great question uh so here it's it's agnostic in the sense that uh you could imagine you could model the the the system that you're trying to model in all in explicit detail so that you know absolutely everything that's going on in that case you would have no

Randomness in that case you still have a stochastic differential equation with no stochastic term which is an ordinary differential equation but still atic differential equation so the same math apply sure you just have a random term that's like zero yeah yeah yeah maybe there are things that you cannot model accurately in the system so you put them in some noise and the same math applies so I would say it's noise from the perspective of the modeler or not and and we don't actually care the point is that once you start from the assumption that you can model the system in that way then then things carry through and I make the presumpt I I'm presuming here that that noise is Gaussian is considered gaussian it's consider a normal distribution gem yeah so generally for uh for for you to be able to do mathematics you take Gaussian noise is that an issue well it depends on who you ask like uh most math and physics is based

on gaussian noise because it represents the most accurate way you have something called the central limit theorem that if you have a lot of like fluctuations they kind of average away in a pile of sand yeah yeah so your noise turns out to be often times very well approximated by gaussian process uh but then there are instances where gaussian noise is not accurate so for example if you're talking to people who model the financial Market they do not use gaussian noise they use noise with like jumps to model like a very sharp um yeah jumps in the market so gaussian noise is a good place to start I think ultimately as we work on the free energy principle mathematically we want to be much more inclusive sure isn't that I mean just as a side isn't that a remarkable fact that like stochastic processes tend towards Gaussian it's just like yeah I think it's awesome it's amazing yeah yeah I mean you have all these remarkable properties of the Gaussian distribution um and it's also like the only one distribution that we actually know how to work with very well so it's a good thing yeah it's cool um and then I guess my other question about the Langevin equation is you have sort of  $f(x)$  as the deterministic force so it's a function of the states of  $X$  the flow of  $X$  is that right yeah right yeah but I that doesn't have a kind of  $\partial/\partial t$  next to it um why is that not uh a function of time as well so it could be in its most general form  $F$  the flow would

be also a function of time now if you think of it let's say you have a state space and you have time you can just incorporate time as another element as another state so you consider your state space to be space time and then you're back at  $F$  of  $x$  right so I think that's why that's why often times people just go with  $f$  of  $x$  cool okay excellent now let's jump to the next step we have density Dynamics and the fuer plank equation and the path integral formulation right very fundamental question what is the difference between those two there there's no difference there's just different ways of articulating the same thing okay so the  $L$  of equation gives you like the rate of change as a function of the flow and a random component random noise yeah uh then when you sort of solve that you you sort of get um this is the probability that the system  $T$  takes this trajectory and this is the probability that gets this other trajectory and so forth this information is what's carried in the path integral formulation the path integral formulation gives you the negative log probability of a path  $y$  and then you the focal plank equation uh which says at any point in time this is the distribution that enclos the system so you might in a predefined place that you know I mean the system might be might start in a particular location in safe space  $m$  because of the flow and the random

fluctuations uh at time  $t + 1$  the system will be it could be here it could be there it could be there with a different probability and then there's a prob distribution that summarizes this and this is what's including the focal plank equation FOC plank they both come to the same conclusion you run the maths both ways and they all converge the answer that it's it's exactly the same thing yeah it's like different ways of looking at the same thing and depending on the question that you want to answer uh taking one formalism or another is easier but the point is that you can always go from one to the other and back nice okay that's cool um okay let's jump into the fu plank equation because I've been I've been doing some reading and I might be wrong but I'm excited about it it's excited me so cool yeah so from what I've learned it's a partial differential equation yeah that rate that relates the rate of change of the probability density function so that's that  $P$  with the little Dot and  $X$  comma  $t$  with respect to time drift and diffusion right how's that sounding yeah it's a great summary I can elaborate on that a little bit to please well people have no idea what I'm talking about so okay well you'll see that you got it perfectly right oh thank God for that all right please explain in a I equation you have the rate of change of the state  $y$  the flow and the random term yes um now this is what physicist called it but if you

talk to mathematician he would say you have the rate of change of state States the drift and a random term which they call the diffusion yes nice or which is characterized by the diffusion so then the focal plank equation is very similar you have instead of having the rate of change of the state you have the rate of change of the probability density that describes the state and then this is given as a function of the drift so the flow and the diffusion so the random term Okay cool so so um if we were going to cast this in like in terms of just cartesian coordinates just X and Y axis would it be right to say that we could have X as like the position let's say of because I I read there's a lot here in sort of brownian particles so let say I had my particles in some you know doing brownian motion the X would be the position of some particle and the Y would be the probability of finding it there yeah when you talk about the focal plank equation yes yeah so the focal plank equation would would be that so at each point in time the the solution to the focal plank equation is the density of the system that evolves over time and so you could plot this on the XY axis as you said on the x axis you would have the position of the system on the y axis you would have the probability probability distribution and so if we imagine things are simple so if we imagine the system is always gaussian distrib according to gaussin distribution then if you solve the FOC

plank equation you would have a gaussian  
that Evol over time Su the mean would be  
shifting and the variance would be  
shifting also potentially nice okay cool  
that's that's really useful so how would  
we then integrate time is it is it this  
like is is the is that X and Y AIS  
considered like a snapshot in time yeah  
okay excellent yes okay cool that's  
actually really useful because I think  
people might got confused by that um  
yeah now let's get to so this gives us  
density

Dynamics but when we talk about active  
we about we talk about steady state  
densities right so people I think have  
this intuitive notion from whether  
they're coming from dynamical systems  
theory or something else of a kind of  
basin right in ATT tractor set I mean I  
think of it kind of like a magnet um  
metaphorically what is how do we get  
from density Dynamics to a steady state  
density um so I should cave at this by  
saying that the latest formulations of  
the free energy principle are not based  
on having a steady state density  
that's like a new thing and I think for  
like intuition it's best to think about  
the steady state case okay  
so um kind of like the intuition is that  
if you're modeling an agent uh it has a  
preferred set of states that it likes to  
come back to or that it like does  
everything it can to like be back at  
those States you could think of like  
homeostatic States for  
instance um you can also think about the

fact that if you're modeling A system that does not dissipate in time so let's say like not a gas for instance uh then the system should also have like a steady state at least over a short period of time because it doesn't dissipate right so now we're going to try to model um I mean the free principle is about modeling or analyzing those systems that have a steady state that have these sort of preferred set of states and so you ask if a system has a preferred set of States and it's also decomposed into an organism its boundary and what's not the organism again this could be like a living organism it could be a stone it could be anything that has a boundary uh then you basically look at what the internal states do what do this what does the boundary States uh do how do they evolve and what do the external states do and then the conclusion is that in general the internal States look as if I mean I mean they they're you can describe them as inferring the external States based blanket States yeah yeah yeah yes that's an interesting formulation and one we could come back to which is this kind of uh yeah tracking in some variational Basi and inference way um we've spoken about this sort of looking like very important two words in the free energy principle um a number of times in this in this podcast but I was curious more about in terms of the fuer plank equation I don't know why

I've turned into a  
um so so this is the way that I would  
think about and please feel free to from  
what I've learned very you know in a  
very shallow way is that we had this um  
this probability density function so  
let's say as you say we have our  $X$  we  
have our position  $y$  we have our  
probability distribution let's just say  
it's gaussian and per attractor State per  
organism per attractor State it's going  
to be different right so mine might be  
really close to this size yours whatever  
um what what I kind of understood as a  
steady state is over time so let's say  
now we like put all of those images of  
the  $X$  and  $Y$  axis on top of each other  
right right and let's say they were  
somewhat  
translucent you wouldn't if it's steady  
state you expect that probability  
distribution to like stay on top of  
itself if it's not you'll expect it to  
shift from side to  
side yeah or or like become completely  
flat all okay all become completely flat  
I guess reincarnation would be kind of  
gaussian flat  
gaussian what what do you mean by  
reincarnation sorry well let's say I it  
was it was a tongue and- cheek comment  
but let like I have a gaussian  
distribution of being in a certain state  
right right right I die I enter into the  
EPA I dissolve into the heat I flatten  
out and then I reappear as a frog and I  
enter into a new gaussian distribution  
yeah that's a good way to put it all

right nice okay so is that is that does  
how does that sound mathematically what  
does that look like mathematically okay  
so um I think one yeah I think there's  
kind of like two  
scenarios uh  
one one is that there's no steady state  
and the system  
dissipates um and therefore you're  
you're like snapshots of uh of the like  
the density at each point in time will  
converge to a flat distribution  
okay um and the other case is that  
there is a steady state and uh the  
snapshots converge onto like a a clear  
well defined distribution yeah which be  
like a gaussian or something else and  
the is that the point the point of a  
steady state is uh that if you start the  
system in that distribution it will  
remain in that distribution sure  
although okay okay fine and and that's  
strict as in you will stay within that  
distribution or is there permission  
let's say in sentient creatures like our  
own for that distribution to change  
somewhat um so that's a good question uh  
there's kind of like two answers to  
that you being at a pre-specified  
distribution does not mean that you're  
not doing interesting things yeah so if  
we just think about a convection  
cell you basically like or or a tornado  
you have like this circling  
around uh the probability distribution  
of where the gas is or where the turning  
is just stays constant it's kind of like  
this ring shape uh but you still have

some interesting motion  
inside um right so I might not  
always doesn't  
me right because I might just you know  
over time it will look like I'm  
converging on that state but I might not  
always be  
there yeah your probability it's kind of  
like hard to wrap your  
heads um if you think about like  
homeostasis right if you model if you  
like let's say temperature if you model  
temperature you can imagine it would  
have a gaussian distribution sort of with  
the Peak at 37.9 degrees Centigrade  
let's say but you are not going to spend  
your entire life at 37.9 degrees  
Centigrade exactly you're going to move  
around and on will be according to this  
Gan distribution right right okay cool  
um okay good um I I want to jump to that  
notion of the internal States tracking  
the external States cuz I think people  
struggle with that um and I definitely  
did for a while so so the way that I  
read internal and external is tracking  
and tracked um that for me has been a  
really parsimonious and useful way I  
think I got that from Maxwell that  
that's a really nice way of thinking  
about it because the internal external  
is such a philosophically Troublesome  
distinction so tracking and track and  
then I think of something like the  
equation variational free  
energy and I think about the KO  
Divergence between the recognition  
density and the true posterior and the

maximization of model

evidence now in

that I guess I'm not I don't have to

necessarily make this strict distinction

between the internal States my States

and the worldly States because all I'm

doing there is I'm saying I have you

know give uh given some observation I

have a probability distribution of some

State whether they're mine or whether

they're the world's because for me I'm

not going for me there that distinction

is not that strong because for me I

think you could pose that actually all

that we are are just this aggregation of

worldly States right I am just I am one

temperature distribution I am one

glucose distribution that's not mine

that's the

world's yeah um so so that's the first

thing right so we're doing this x uh X

give and Y this recognition density

and then you have the maximization of

the model evidence which is the

probability of why given your

model but again there's nothing like

internal that's going on there it's just

statistics so do you that do do do you

think that it's somewhat Troublesome to

cast like put this little homunculus or

this little ego in the internal States

and being like he's doing sufficient

statistics is that do you see that

slightly

Troublesome um I think it's a nice

picture for ination but it's

certainly uh it can be misleading for

the reasons that you pointed out uh

internal and external States can be  
interchanged yeah and again it's all  
about the blanket yeah the external  
external States could be part of your  
own body and the internal States could  
be part of your brain and the blanket  
somewhere in between or the internal  
States could be like a school of fish  
and the external States could be like a  
predator right um yeah I just never  
thought I this is no attack on anyone I  
don't know who came up with it probably  
KL sorry KL I never thought that was a  
particularly good formulation of the  
free energy principle I always thought  
that um you know things that seem to  
persist look like they're maximizing  
model evidence for themselves is way  
better um but this is something to take  
out with him yeah but but it's kind of  
like two things we're touching on here  
because tracking and the Mel  
evidence uh they kind of like two things  
are going  
on because when you think about it the  
free energy is this  $K$  Divergence plus  
surprise uh and we use this tracking to  
like minimize the free energy right so  
what I mean by that is the maximization  
of model evidence or I think it's the  
minimization of log model evidence but  
it's the same thing  
um that entails that in of itself what  
the way I'm trying to frame it is that  
that entails some form of tracking yeah  
because you've got that you got that  
probability of why but it's sort of  
given your

model um so yeah I I just think for people this is again this more an education point about active inference not about the maths or anything um I just sense that for me it's always been more intuitive to think about that although again it's hardly intuitive is it I mean it takes a bit of time to really wrap your head around yeah there's there's another important point that came out in the research uh when we are talking about the fact that you can I mean a lot of people say that you can always interchange uh internal and external States and that one is tracking and the other is tracked but it turns out that this is not always the case because for the internal states to track the external States they need to have enough like representational capacity make inferences about all the internal states are out there they don't have enough representational capacity maybe they might just infer some or infer linear combinations of external States in any case they're just going to have like part of the picture okay that's no that's that's interesting which means that for external states to track internal States they also need to have like enough representational capacity and a sufficiently high quality coupling with their own sensory states with the boundary to make inferences about the internal States so you always have this synchronization and this inference going on you Cann necessarily say that the

internal States will be tracking all the internal States all the time or or vice versa right kind of depends on the Integrity of the coupling and how much representational capacity these states have well again that's why for me this maximization of model evidence just seems a little bit simpler because whether you're a drop of oil or a rock or human we can sort of arbit you know not arbitrarily but we can generously give out models for each one of them and say that they look like they're maximizing model evidence yeah now um like how long something lasts is a different question um so my you know the the the salt that I put in water maximizes model evidence for itself being so for a very short amount of time so you know you could cast that as it surprise will being very high but at least for while it's salt it's maximizing model evidence and then I guess it's maximizing model evidence within a Markov blanket that's the water salt solution anyway it's an interesting foray but it's no that's a good point yeah we're now going to jump to the next part of this so people who want to follow along by the way a lot of this is from the free energy principle made simple but not too simple or I think that's the name of the title that's right yeah that's a great this this is a well I wouldn't say it's necessarily a perfect starting point for someone who hasn't done maths or physics don't we I would I would recommend um the

textbook so paret Zulo and friston you  
got recommendations actually yeah  
there's also Maxwell's paper on Basin  
mechanics yeah that's very good and off  
beliefs and this is really cutting edge  
in describing the math but it's also  
meant to be much more  
approachable yes okay that's yep so  
Maxwell's 2023 paper which is on the uh  
pre-print server um the textbook and  
I've also been reading and we'll come to  
this I've been reading sanjie Nam josi's  
draft of his p uh book it's it's it's  
being sort of distributed there's a  
group in the active inference Institute  
where you can read it and that's been  
really useful so there's stuff out there  
and and there podcast  
and so yeah but if you wanted to follow  
along with this this is that uh B uh  
free energy principle made simpler but  
not too simple so the next thing that  
you you come to there is solenoidal and  
gradient  
flows right so maybe we can talk about  
maybe we should start by just explaining  
what those two flows mean and how  
they're distinct from one  
another okay um so this is a the  
composition of the flow in the Lan  
equation yeah and  
so uh to put it simp to put it very  
simply  
the um well just to give an analogy in  
Vector calculus you have this  
fundamental theorem about that's known  
as the hamle dec composition yeah which  
says that if you have a vector field so

think about a bunch of vectors sitting on a table uh you can decompose them onto uh one that's kind of like circling around an origin to put it simply and one that's going inside or outside okay so mathematically this is what we're doing here to decompose uh the flow in the L equation now what this really means in practice is that you have uh the solenoidal flow which is the interesting part of the flow and the how did you call the other one uh what did I call it gradient flow right and there's a gradient flow that's the in uninteresting one if you are um if you are a living system you have by definition solenoidal flow and why is that solenoidal flow is is um related to time irreversibility and entropy production um okay if if you don't have solenoidal flow it means that the system if you if you play it in time uh as time goes forward and then you you reverse the movie statistically the system will look the same so it's essentially a dead system it could be like a stone for instance yeah on the other hand if I you know you play the podcast for 20 seconds as time goes forward and then you play it backward for 20 seconds it will look super weird and very different right this is the proof that there is time irreversibility time has a there is a direction time um and this is yeah characteristic of the fact that we are alive interesting okay so let me just get some some more you know some simple

things outlined here so you said the hel  
holtan the Helm Holt's decomposition is  
of the flow is that if I'm looking at  
the long Jaan equation that's not the  
that doesn't include the random noise  
that's just  $f \circ f$  yeah it's a de  
composition of  $f \circ f$  but also done in  
relation to the noise term so is the  
noise term so would that what I'm asking  
here is when I'm splitting it into  
solenoidal and gradient is that  
splitting just the  $F \circ f$  or the  $F \circ f$   
and the  
yeah Okay cool

so okay so there's a there's a  
dissipative aspect in all of this so the  
analogy that I thought of was um if you  
create a Whirlpool let's say in your  
sink or in a cup of coffee you get this  
solenoidal flow right it's conservative  
but due to I presume just friction you  
get this dissipative uh ele this this  
aspect where the swirling motion Fades  
away yeah totally right let's is that is  
that the gradient flow then sort of  
infiltrating a purely solenoidal flow  
yeah that's absolutely right another  
example is if you put milk in your  
coffee yeah there will be like the  
standard  
mixing uh of things which will be the  
gradient flow then if should start  
stirring the cup of coffee with a spoon  
then you're adding C flow and is it  
where they meet that Things become like  
we can say things are dispa of because  
for me philosophically dispa of means  
that you had order and now you've got

less order but if you if everything  
is if there's no solenoidal flows then  
arguably there's no is there is there  
any order to begin  
with it's hard to say like I have  
yeah I have trouble relating these  
Notions to  
order but I think there's like there's  
like an indirect way of relating them  
which is so Noto flow is always  
something that circles around something  
yeah you think about the Circadian  
rhythm yeah us waking up and going to  
sleep all these are Cycles they are  
examples of sidal flow an aof flow has  
um produces entropy so I think that's in  
some sense related to order and disorder  
if you want to maintain as a living  
being if you want to maintain tidal flow  
or just as a physical system you need to  
be using energy and dissipating entropy  
so that's kind of the  
idea and that's just a consequence of  
the mathematics and the physics yeah  
yeah basically okay the fact that we  
have senoidal flow within us like  
rhythms and Cycles means that we need to  
eat and uh and secrete right right okay  
okay because in some sense the that's a  
form of order which is neg  
entropic yeah yeah okay now uh okay so  
what is this um the other thing that  
comes up is the term  
Divergence right so people say that  
solenoidal flows are zero  
Divergent yeah what does that what does  
that mean  
right uh it's kind of that's just kind

of a mathematical term uh but uh because  
if you take the Divergence of the vector  
field it's zero but that's getting to  
that I'll give you a picture if we if we  
are at our steady state distribution so  
let's say um our system is two  
dimensional and it's distributed  
according to gaussian that's sitting on  
my table  
here um the so-called flow is uh motion  
around the Contours of this gaussian  
distribution and the gradient flow is  
motion towards the peak or away from the  
peak interesting and so we have our  
overall motion which is given by the  
flow in the Langevin equation which could be  
in any direction and then um the fact is  
you can decompose it on motion towards  
the peak so towards states are more  
expected or preferred and motion away  
from the peak strictly away from the  
peak these two are Divergence free flow  
you can also the other part of the  
decomposition is motion that just goes  
around the Contours of the distribution  
which means that it does not approach  
the peak or go away from the peak is it  
if I'm thinking about it  
diagrammatically does it matter where  
that circulation is  
going does it so so it could be at the  
base of the gaussian which means it would  
in some like again just diagrammatically  
means it would be wider or it could be  
at the top of the Gaussian which means it's  
occupying these high probability  
states does it  
matter um I think it matters as a

signature of the system probably I mean  
this idea of Coto flow is one of the  
most important like signatures of uh of  
being alive and what your states are  
doing what kind of Cycles you have in  
your system or in yourself uh so it's a  
very important um descriptor of what's  
going on okay cool and the other thing  
was um and again you can tell me when  
I'm when I make a terrible fuxa is that  
solenoidal flows break what's called  
detailed balance yeah and what I've  
learned about this is that detail  
balance is when the rate of transitions  
of any two states is equal in both  
directions that's right so like you can  
think of it in terms of I I I thought of  
it in terms of like a crowd going in and  
out of a  
like a stadium or in and out of a room  
that rate is going to be the same in and  
out and that I believe and again you can  
tell me if I'm wrong there's something  
to do with boltzman boltzman's h  
theorem which says that that that  
detailed balance implies zero in that  
entropy right um so so two parts um it's  
absolutely right that detail balance is  
means that there is flow  
um in and out say of a stadium at at the  
same rate it means that if you play the  
flow in and out of the stadium forward  
or backward it will look the same  
there's no production of entropy and  
there's no solal flow and the system is  
time reversible yeah yeah yeah okay so  
that kind of explains why the it's sort  
of N Net Zero entropy right if entropy

is foundational to the passage of time  
yeah yeah yeah basically yeah as soon as  
you have S flow time uh starts taking  
place you start having a narrow time  
okay so if if we  
took so yeah okay so this is interesting  
so let let's let's kind of this is an  
odd exercise but we can kind of build a  
human from scratch right  
so let's try right um so the the  
gradient flows are going up or down the  
probability distribution so we take a  
human as just our gaussian distribution  
it's kind of going up down but let's say  
it without any solenoidal flows we have  
this detailed balance yeah so it's going  
up and down  
now does that not mean that because it's  
doing it uh equally you're going to end  
up at that  
Peak uh no because you have equal flow  
towards the peak or away from the peak  
so how does that average out what does  
that look like in terms of where I where  
the crowd might be or they're just  
everywhere  
are we talking about the human or or the  
crowd going in and out of let's say the  
crowd let's say because we've got  
detailed balance so let's say the crowd  
okay okay if I'm gonna say okay uh I  
have a random friend X and they're they  
belong to this crowd that's uh behaving  
in terms of detailed balance and I want  
to find him is there a probability  
distribution that could help  
me to find your friend yes who's going  
in you know is part of that TI

um yeah I'm not sure about that okay  
what I will just say is that like if you  
have one exit in your stadium with  
people going in and out at the same rate  
yeah and they're kind of like mixing  
with each other so who is who you have  
detail balance you just having one  
entrance and one exit then you do not  
have detail balance because you have  
people coming in one way and going out  
the other way so you start having a  
cycle

ah okay now that's that's okay that's  
really useful nice so let's now talk  
about the human in a sense because the  
hum a

combination of we're kind of what Carl  
would say we're in this goldilock zone  
so we're not orbital planets with just  
solenoidal flows so we so we're not just  
this cular uh circular thing but we're  
also not just NE entrop non-- entropy  
one it right so we have both so we have  
both yeah so yeah it's a very  
interesting sweet spot if you just  
consider systems that have just the  
gradient flow yeah are dead and you know  
they might look like gases and stones  
and stuff like that yeah yeah yeah if  
you just look at things that has tornado  
flow well I would say they're probably  
dead as well it could be like like  
planets orbiting around the Sun like  
they're going One Direction and not the  
other and we are kind of like in between  
where we have both going on so I guess  
this is where my question was going  
which is okay let's say let's take a

Rock The Rock has uh just gradient flows  
so it's dead but it seems to me that the  
rock is doing a pretty good job at self-  
evidencing right  
so does in some ways the solenoidal flow  
take us away from Yes it gives us  
animacy and whatever but does it take us  
away from like just being being able to  
rigidly self-evidence rigidly maximize  
model evidence no um no no it does not  
and so the beauty of the mathematics  
here is that we're we're taking the most  
General sort of equation that can  
describe the world so the F principle  
applies if there is C Noto flow and no  
decied flow yeah it applies there is  
decied flow and no solenoidal flow and  
it applies if there's  
both yes but what I mean by this is it's  
it's probably worth mentioning that I'm  
kind of thinking about this over time so  
my uh my plastic bottle is going to  
outlive me and and because things over  
time have to you know have to abide by  
the free energy principle they have to  
look as if they're maximizing model  
evidence or doing some variational Basi  
and inference you could say if we had  
some normative sense which again I'm  
quite skeptical but you could say like  
the water bottle is doing better  
self-evidence over a period of time  
uh yeah over over a long period of time  
certainly and that's because well that  
seems to me to map on quite nicely to  
solenoidal gradient flow distinction  
because like if rocks and uh I don't  
know particles are not

solenoidal where where am I I feel like  
I feel like there's something I'm  
missing yeah that's uh I mean I think  
it's very interesting you point out that  
things

that do not have tidal flow sort of  
endure over a longer period of time um I  
never thought about that okay okay fine  
fine fine fine I won't push it then okay  
that's really okay yeah it's interesting  
I need to think about it yeah yeah yeah  
it well God don't know what kind of  
worms I've just opened up I really don't  
want to get shouted out on the comments  
um okay so something I do understand a  
little bit better is this uh KO

Divergence uh that's more up my streaks  
I feel like it's a little bit more  
cognitive sciencey so it's coolback Lio  
Divergence and and as I said this Um  
this can be cast as a distinction if  
we're doing Bayesian inference between a  
true posterior and what's called the  
recognition density um which is like  
your version of your of of a posterior  
now there was something you brought up in  
that uh or car Bop or whoever wrote it  
in that um simplifying the free energy  
principle paper which is called a Fisher  
information metric tensor which measures  
changes in the K

Divergence it might be so people I think  
view the variational free energy as a  
kind of snapshot in time so given like  
this perceptual inference how big is my  
K Divergence given my action how much  
surprise is there but this seems to me  
to integrate change over time so what is

the Fisher information metric tensor and how can we integrate that into a more accurate or more faithful model of the world or model of ourselves right uh so once you accept that you have a recognition density about the world and that this is minimizing free energy you can ask how much does your recognition density or your beliefs about the world have to change in order to minimize free energy and so if you if you want to quantify how much your beliefs move in a certain amount of time you need what's called uh an information metric information metric in the sense that it uh quantifies how much information changes over time the information have about and the information the fishal information metric is what allows you to do that nice okay cool are you are you familiar with this notion that's been coming out of maybe like the more philosophical or cognitive science side of active inference so people like Mark Miller and Julian Kine of prediction error

Dynamics um I mean intuitively I can see what that would mean but yeah the idea is that we have like a higher order prior that we will track how good we're doing at minimizing free energy over time and this constitutes and and this in terms of psychology maps onto something like positive or negative affect because you know we've got a great we're doing well right we're doing well we're finding these slopes of um Vari variational for energy that we can

descend down and there kind of an explanation for why we don't seek out dark rooms so there's one other solution to the dark room problem it sounds quite similar to what you're saying which is except what you're saying seems strictly I I don't know in terms of the K Divergence tied to perceptual inference but we can say that a system is becoming better perceptual inference because over time those KO divergences are going down is that reasonable to say well yes and no because if the data stream sort of like stayed constant oh yes or if you stop the data stream then your K Divergence would go down because you would kind of like get a free energy minimum yeah but always have new data coming in so you have two forces at play you have you uh who are like predicting the data and minimizing free energy but then you have data that's potentially increasing the free energy right but is the world but going back to our long equation the world is the world is modelable in some sense it's predictable in certain senses so I I may be wrong but it seems to me that the more data you get over time the better your model is going to be right it's it's just going to become more powerful right to use a statistical term what what's the kind of so that seems to me that like your information metric should be pointing that you know your K divergencies are going down even if you're getting more data because that data's just it's got a similar

distribution to the data that came  
before right okay so so I see I see  
where where the confusion lies  
um I agree with you that as we get more  
data our model gets better and our  
free energy um in general tends to be  
lower so that's one truth the other  
truth is that even though even even  
though I mean that's a fact your the  
data that's coming in keeps  
changing uh which means that the  
posterior density the true the true  
keeps changing so your recognition  
Dynamics always have to keep tracking  
this posterior distribution right now  
the fish looks at how much these um  
recognition Dynamics have to travel in a  
certain period of time to match the true  
posterior is there an equation that  
governs that because obviously I've  
become obsessed with equations um is  
there is there how much your beliefs  
change no not just how much your beliefs  
change but kind of how the data  
changes how the data changes that's the  
longine equation long equation but it's  
so that's the function of the  $X$  being  
the worldly States  
yeah got it got it okay okay yes okay I  
see where my confusion was I guess I was  
viewing the world  
as kind of static I mean you can even  
like take away the  $r$  the the random  
noise right if your  $f$  of  $x$  is changing  
your agent who's modeling that is going  
to have to keep updating keep changing  
their model nice nice okay cool that's  
that's that's very useful okay excellent

um okay have I got any more maths um I got a bit more maths I guess one one something I wanted to ask about is this is a broader question so in so in active inference we have discrete State spaces and continuous State spaces so and it's kind of about the how we carve up time in some sense so continuous State spaces is where it doesn't really make sense to put time into separate chunks or sequential chunks so you know if you're thinking about attention or perception or perception let's say um I know perception includes some action but let's just take perception it's better to kind of model that in terms of a continuous State space if you're thinking about decision- making you would think about that in terms of discrete State spaces because an action policy is a discrete thing in discrete state space we've seen this kind of proliferation of So-Cal pom DP schema um so these are these partially observable Markov decision blank but process close always think about Marco blankets um is there an equivalent of that in continuous State spaces uh so you can definitely have continuous uh partially observable Markov decision processes um so so let me take a step back there's two things that play there's a whether the state space is discrete or continuous and whether time is discrete or continuous okay now in active inference implementation so far you basically have either continuous

space continuous time or discret space  
discret time yeah yeah or both but  
people haven't really been playing with  
continuous space discrete time right uh  
continuous time discrete  
space but there's no reason why in  
principle you you would not be able to  
do that okay that's cool well uh this is  
again my very much lay and knowledge of  
physics um do we not take time and space  
to be somewhat  
interwoven  
um yeah this is a bit beyond me I guess  
I guess I guess a way around this is to  
say that we're modeling right so in our  
modeling we can take time as continuous  
and space as discret and vice versa yeah  
this is this is only about modeling  
about how you represent the world but  
there is a a deep uh a deeper question  
which is how do we ultimately like  
represent the world in physics and it's  
true that some theories you know  
consider time and space jointly but for  
us humans uh yeah we don't need to get  
into that for active res modeling okay  
so we we've we've kind of clarified that  
so when so where would the pomdp schemes  
be able to be used if it's not just  
discrete time discrete space they  
applicable to continuous space  
continuous  
time so so where would you actually use  
them so I've seen for example just to  
give some background I've kind of seen  
pom DP schemes used in mainly just  
aspect like context of decision  
making so um for example yeah going back

to the flow paper flow involves  
selecting action policies meditation  
involves mental action um attention  
right like um  
intentional uh attention uh endogenous  
attention is a  
decision these can all be modeled quite  
nicely but to my eyes everyone who talks  
about perception will use something like  
a predictive coding scheme yeah um is  
there is that I'm just curious about  
what is if there is an alternative to  
that in the  
literature in the literature no okay I  
was talking to Ryan Smith recently and  
he was interested in continuous State P  
DP with  
discret and so the idea here is that you  
can Model A continuous representation of  
the world and a sequential decision  
making at the same time of that  
so possibilities okay cool I'm speaking  
to Ryan fairly soonish so that that  
would be useful that would be cool  
that's a good thing to bring up yeah  
that's really interesting I don't know  
this brings to mind there's this  
incredible paper by a guy called Juan  
Diego botar and I think his name is  
Zacharia  
Jabara and it's sort of symphasis  
subjective time and objective time  
through a basian model and I'm going to  
tell you I stared to that graph for  
about 6  
hours um and it still did make sense so  
that's a you know if anyone wants to  
read that and try and think about how

maybe not space and time but like two  
different types of time can be coals  
that's a good place to start so I I'll  
give you another theory of mine uh I  
attended this wonderful work workshop on  
computational neurop  
phenomenology uh last year and I had a  
wonderful conversation with annel Seth  
and we kind of converged onto the  
hypothesis that subjective time is uh  
the amount of distance that your beliefs  
travel over some period of objective  
time okay cool so here here's the thing  
like when you when your beliefs move a  
lot yeah and you can quantify that with  
the Fisher information metric you would  
have a perception that a lot of time has  
elapsed okay so you've now got me on  
this new pet hobby of mine because  
I'm sure you haven't seen it but  
literally yesterday I published a  
preprint called distrusting the policy  
how inference over action shapes our  
perception of  
time that's cool I should read so so  
please I would love some feedback  
basically I wrote it so and this this  
started off as being trying to explain  
temporality in flow States and I thought  
no actually this works I think this  
works more  
broadly so I'm not going to spoil it  
well I will spoil it but basically my  
idea so I'm building off what yakob  
Harry called distrusting the present so  
this is very similar to what you're  
saying so yakob Harry has this idea that  
this the flow of time is contingent on

um the rate at which we update our  
perceptual hypothesis about the world so  
if I'm in a volatile environment I the  
uh the state estimation I have at time  $T$   
is not going to work for  $t$  plus one and  
so what happens is the moment I find  $T$  I  
immediately start discounting it which  
you can consider sort of precision  
waiting and immediately I have to update  
that and he C that now I I don't think  
that that totally maps onto the  
phenomenology because I'm really curious  
about what you think about this and  
again I will try and do it quickly  
because I don't want to bore you but we  
have  
um so I say so he was kind of it seems  
to me he was rooting uh flow of time in  
predictability but the problem is is if  
you're going to cast it only in terms of  
perception or only in terms of the a  
matrix then you you've got a slight  
problem because there are two different  
types of predictability you've got one  
which I call static predictability which  
is basically when you're staring at a  
wall now when you're staring at a wall  
your perceptual hypothesis is going to  
be pretty good let's say over a prolong  
period of time and so Jacob's Theory  
says time will slow down and it does  
right you stare at a wall for an hour  
it's going to feel like a day yeah but  
then the other problem is something  
called Dynam what I call Dynamic  
predictability so let's say you've  
played a video game level once it's very  
exciting time seems to flow quickly now

let's say I make you play that same  
video game a 100

times now perceptually the world is  
changing you got to do all of that state  
like that state estimation that  
combination of the D Matrix and the a  
matrix needs to keep happening at the  
same rate and I want to flesh this out  
experimentally but I have a hunch that  
time is going to move way  
slower

so what I've what I've now said is it's  
about the frequency with which you  
update your action policies and the  
reason why I said that is because with  
each action policy updating you also  
have to update your beliefs in the C  
Matrix and the C Matrix is your expected  
pre uh your preferred sensory  
observations and that I think accounts  
for the perceptual aspect of the flow of  
time that's really that's a really  
interesting hypothesis yeah yeah so I so  
it's Friday the 19th of January on  
Tuesday people are not going to be able  
to you know this is not people can't go  
back in time anyway I'm doing a talk  
about it at the at the institute on  
Tuesday um so anyway that that's so I  
think we're converging on similar points  
here I think it's kind of where is the  
source of your information change so for  
me it's the C Matrix really because it  
can't be the B Matrix because in Dynamic  
predictable so uh B Matrix of State  
transitions in dynamically predictable  
situations like playing the same video  
game 100 times I'm going to have the

same number of B  
matrices compared to when I played it  
the first time so I don't think it's the  
B Matrix but because I you have to keep  
doing this expected free energy  
calculation for each action policy and  
the C Matrix feeds into that my hunch is  
that it might be rooted than that right  
you you could also think about the fact  
that if you replay the video game uh  
like many times mhm your belief still  
have to move to adapt to the current uh  
situation in the video game but there  
will be less and less novelty yeah  
because you adapted your like big world  
model to like predict the video game  
very so how that translates to in  
practice is that you have lowlevel  
beliefs that that move a lot to you  
adapt to current situation but high  
level beliefs that don't move because  
okay nice everything again I think I'm  
trying to get this to be done  
experimentally to  
see if it's if if this Theory maps onto  
anything yeah it would be really nice to  
test but that's that's a very  
interesting yeah I'm always in favor of  
all all of the empirical  
experiments I feel as a community we  
we're lacking that I think that's right  
and I think it shouldn't be that hard to  
do I think you just have to give people  
statically predictable context  
dynamically predictable context and flow  
context yeah I  
mean um as soon as you can map a p onto  
the data that's coming in be very easy

to model everything that's going on well  
I may reach out to you or LS on that one  
um Okay cool so that that was a fun  
little for um okay actually let's let's  
um we we spoke we very briefly touched  
upon

attention um and I have a question about  
attention if I may i' I've been I've  
been I've been grappling with attention  
for quite a long time um so attention is  
kind of construed as optimization of the  
likelihood distribution in active  
inference of the a matrix in the pomdp  
schema

now there are these really lovely papers  
by his name is mitah so uh mital 2016  
2018 and 2019 I think and he shows he  
gives this example of like searching for  
a red pen and he says that when you're  
searching for a red pen what you do is  
you downweight sensory Precision of  
stimuli that are Rel irrelevant to that  
context and so basically you end up  
doing Salient Behavior which is we can  
view as epistemically Salient because it  
yields Information Gain but the  
information that you  
want um now given that um yeah so that  
makes sense because task relevant  
stimuli are downweighted so only task  
relevant stimuli involved in this maximal  
Information Gain okay fine but  
pragmatically so so that epis that  
information gain is like I didn't know  
where the red pen was now I know where  
the red pen is yeah but in terms of  
pragmatic in terms of pragmatics we know  
that action policies in uh Act of

influence are driven by a kind of almost  
a false belief a false preference so I  
might say I have a red pen and I'm going  
to go out there to confirm that belief  
because without it there's lots of  
prediction error or variation or free  
energy this seems to be a kind of issue  
because on one hand I'm gaining  
information about the world so I kind of  
knew that I didn't have a pen and now I  
know where the pen is and so my I have  
this K Divergence but then I also have  
this oddly pragmatic belief that I do  
have a red pen because without that red  
without that belief I wouldn't be  
motivated to go and get the red pen so  
I'm I'm trying to Grapple with how to  
resolve that clash and I'm curious if  
you can  
help yeah not sure um but what I'm  
thinking is uh I always think about  
attention as control of the a Matrix so  
kind of like in large large's paper on  
meditation mhm uh you make some some uh  
sensory modalities more  
precise and others less precise yeah and  
it feeds into into your  
computations yes  
so yeah see I've always I've always  
found it quite useful to distinguish  
Precision from Precision waiting so  
Precision I just view as the kind of  
inverse entropy or of a  
distribution but Precision waiting is  
actually what I do right I wait  
precision and that's not all going to be  
down weirdly Precision waiting is not  
all down to

Precision well isn't it I think it often is it often is in terms of so they talk about how attention is basically one definition of it is the capacity to kind of sift through the data that you currently have for the most uh informationally reliable or precise um data right so the one that's going to say okay given this I know exactly what caused it but clearly like for example in my searching of the red pen there's going to be loads of things in the environment that are that have a very uh you know very strict uh relationship mapping between the observation and the state so while I'm scanning I might see something and go oh obviously that's my phone right it's not ambiguous whether it's my phone or not but because I'm but the whole point of this Mit paper is I don't I don't care so I'm downweighting that stimuli but that's nothing to do with how well it maps onto some

State yeah

yeah it isn't okay so if I understand correctly this is um because you you want to conform to your prior belief and therefore if your prior belief is very precise mhm uh you will discount whatever sensory information comes in right so my prior belief is yeah so my prior belief is something like I have a red pen and and and so that's very precise and yeah so that leads you to Discount things that are because you're you're yeah you're only going to

pay attention to things that are task relevant yeah then so but that but that's kind of but the point is that's kind of a belief saying I have a pen but then there still is this epistemic gain which is at the heart of Sal which is I didn't have a pen or I didn't know where the pen was and now I do know where the pen is and so you're starting with I don't know where the pen is but for your act to Trigger action you have the belief I know where the pen is to to Trigger action to look for the pen I'm I'm getting I'm getting a little bit lost okay so so um preferences underly action right so a see Matrix so I have the preference about the sensory information that I expect to receive mhm so let's say and and preferences can be casted as just beliefs so I believe so for example uh in some very it's it's some very subpersonal way I believe that uh I'm going to be at 37.9 degrees and so I act so I remain around that now I want to find my red pen we can cast that valuation as a preference as a belief yeah I know where my red pen is and the reason why you would do that is because well the world is not giving you that information and so you've got prediction error and you want to minimize that prediction error so that triggers action to confirm the belief that you have a red pen okay that's on one side but on the other side what captures attention is that which is most Salient and what salience

can be cast as is epistemic gain

Information

Gain and that for me in this red pen

example looks like well I didn't know

where the red pen was and now I do know

where the red pen is

but what you've got on one hand is on

the epistemic side you've start with the

belief I don't know where the pen is

because if I did know where the pen is

finding the pen would not be very useful

right so that's um inhibition of return

that kind of explains inhibition of

return that's on the epistemic side on

the pragmatic side I have I have a

pen so I'm just asking how those two

things come

together I'm I'm so sorry Darius I think

my brain is fried by that's all right

that's right I'm sure someone in the

comments will have an answer yeah I kind

of feel feel that KL will just have an

answer um I mean I'm sure I'm sure he

has an answer to everything Maybe not

maybe not okay let's um a research paper

yeah maybe I I know that one's going to

take me down some wormholes

um so okay last thing I guess I want to

talk about is you went on the machine

learning Street Talk podcast and you

spoke a lot about core knowledge which I

thought was really interesting I really

didn't expect that to come up um but I

know it's one of your interest

terms of what are the kind of priors

that need that are uilt that

facilitate intelligence sentient

Behavior

so I guess what are your current thoughts on how so it's interesting because I actually come from a Linguistics background as I told you before we started and at least you know at Oxford where I was they they love Chomsky so Chomsky has this notion of universal grammar there's something built into the genetic code that facilitates language learning so I'm very familiar with these arguments Poverty of stimulus and these arguments in terms of nativism what are your thoughts on how fleshed out those priors are um in terms of being innate well I think there's definitely innate priors when we were born and this is all studied um I think one of the best people is Elizabeth Spelke y her whole career uh describing and characterizing those priors MH uh these seem to be incredibly precise uh much like we cannot change them and they seem to be super General in the sense that they would apply to any kind of naturalistic or physical environment that you could ever be bored in so these are things like object permanence two object cannot uh interact at a distance but two people can interact at a distance so so two agents can uh I think there's a core knowledge system for um numbers learning numbers there might also be a core knowledge system for language uh they might also be one for vision uh so there's six of them and she wrote a book about it uh which is what

babies know uh that I absolutely recommend with with all of that research so it's a big research program to actually being able to understand those and reveal those coronal systems it's another one to be able to reverse engineer them

interesting yeah I think one of the issues I guess is at least um in the history of linguistics one of the great problems was how rich are these priors so this is what later got known as Darwin's problem uh because tumpsky has this idea that well in the 80s he had this principles and parameters idea which is that what ug is is a bunch of parameters that can be switched on on uh you know given sensory data those parameters get switched on or off and then the problem is you had this massive proliferation of parameters so there were 500 parameters and Fin and finished people you know there was one only for finish but obviously we all had it but only unless you were exposed to finish did you ever turn it on and it just became completely inexplicable given Evolution because how on Earth did one mutation or a couple of mutations trigger that so he now thinks that there's something a lot more paired down something like recursion and I think why why that's interesting is because recursion is not is not what we call propositional Knowledge it's not knowledge that something so Trump might say oh okay you know in the 60s might said okay I know the baby knows that

after this comes this or something but  
now this is just actually a mechanism so  
what we might call procedural knowledge  
when you think about uh core knowledge  
for maths or call knowledge for vision  
or I guess Vision really won't be  
propositional call knowledge for  
language  
are you construing these as  
propositional or  
procedural um  
so well these would be priories in the G  
of  
model which could be  
expressed um in a number of ways it  
could be uh sort of like um like a  
grammar of uh like core components that  
you already have in your G model well  
well let let me put it in a different  
way like when we're born we have a very  
simple gen model and gradually we learn  
about the RO around this and we make it  
more complex now what we have at the  
beginning is is this core knowledge it's  
kind of like a starting point Gant model  
that allows you to right by structure  
learning learn about everything else now  
the structure of that is uh very debated  
among computational people so for  
example Josh tenenbom is pursuing this  
idea of probabilistic  
programs which are B based on symbolic  
grammars and probabilistic rules between  
different  
entities then there's other people who  
might uh model these things with neural  
networks right so in those cases you  
don't have symbols you don't have that

grammar going on yeah it's very much an open area of research cool yeah I mean I guess all these neural networks still have some priors um in in a lot of cases yeah awesome yeah yeah I used to love these debates between Chomsky and people like um Jeffrey Elman sort of mid 1990s proper hardcore stuff cool and yeah I I it's interesting you mention Vision because I always thought that for me the one glaring one where there needs to be PRI is Vision vision is perception is well visual perception is such an amazing capacity yeah to be able to there is a startup called a Common Sense machines MH has pushed the core knowledge of vision really to state-of-the-art level where they have a world model for vision and then they do inference on that world model and they they're able to reconstruct 3D scenes from 2D images wow in in a pretty amazing way so yeah out have you seen these things uh the this is slightly different but you have these neuron Nets that are trained on like uh brain Imaging data and they can take an MRI of the brain and say that this person was thinking about a dolphin or something they can have an image of what that person is thinking about obviously we're given the training that you know there is this I'm not going to say causation correlation I won't go into it but that you know when this part when this brain data is like this this person is thinking this and so

you have the testimonies of the people  
but it just shows the power of those and  
the and the kind of nuance of those  
large language models that they can be  
able to do that um but but actually I  
think it's arguably less exciting than  
people think it is because they forget  
about the training process that  
underlies it yeah and it doesn't allow  
you to understand it just allows you to  
predict exactly exactly and that's where  
that's where our our favorite topic of  
active inference might might come in  
maybe and yield some artificial  
understanding hopefully hope well people  
are getting very excited getting very  
excited L that was that was awesome I  
feel like I've leared I feel like I've  
gone to University again yeah me too  
no this no this was this was so much fun  
I knew it would be fun thank you for  
being gentle with me and kind with the  
maths well thanks for having me no it  
was an absolute pleasure um I really  
want to see what comes out of this stuff  
especially with annel about the Fisher  
information metric and time yeah is g to  
come out really cool yeah cool sounds  
like we're converging potentially on a  
similar idea so yeah it would be really  
nice to talk I need to flesh out these  
ideas and then yeah we can talk more  
Splendid okay well thank you from longan  
and [ ] PL and all of this stuff it's  
been an absolute pleasure so thank you  
for myself and The Institute it's been a  
great pleasure for me too thanks